



SEGURIDAD DE LA INFORMACIÓN

UNIVERSIDAD
POLITÉCNICA
DE MADRID

Parcial #1

5/3/2024



Family Name	Name
-------------	------

Part 1 of 2 : Diffie-Hellman key exchange (4 marks)

Michael Schofield and Lincoln Burrows are two inmates at Fox River Penitentiary who have decided to escape. To plan their escape safely, they have decided to exchange messages confidentially using a symmetric key encryption algorithm.

- a) The plan devised by Michael and Lincoln requires overall 7 inmates to succeed (including them). How many keys will the group need to generate if they want all be able to speak with each other without anyone else being able to access their communications? (0,5 mk)

$$Nk = (n*(n-1))/2 = 21, n=7$$

Michael and Lincoln decide they want to use Diffie-Hellman to agree on their keys. They pick a prime $p=19$ and a primitive root of 19, $\alpha=3$.

- b) If Michael picks 8 as his private value, calculate the value he would send to Lincoln (0,5 mk)

$$M = 3^8 \bmod 19 = 6$$

- c) If Lincoln picks 5 as his private value, calculate the value he would send to Michael (0,5 mk)

$$L = 3^5 \bmod 19 = 15$$

- d) Calculate the obtained shared key from the point of view of Lincoln (1 mk)

$$K = 6^5 \bmod 17 \alpha = 5$$

- e) Imagine Michael and Lincoln make a mistake and pick $\alpha=7$ which is not a primitive root of 19. How would that affect security of the Diffie-Hellman key exchange? (0,5 mk)

If they pick an α which is not a private root of 19, it will be easier for an attacker to perform a brute force attack and obtain the shared keys as there would be exponentiations with different exponents resulting in the same value.

- f) Imagine that the escape plan devised by Michael and Lincoln requires 20 inmates to work. How would that affect the keys they all obtain via the Diffie-Hellman key exchange? (1 mk)

If the number of keys is larger than p , there will be at least two shared keys that are the same as there will be secret values that are repeated. In this case there will be many more than 20

keys so many keys will be repeated within the group which would be an issue for confident.

Part 2 of 2: RSA (2 puntos)

Alice wants to use RSA to send Bob a message that contains the key combination to open a security deposit box that is being stored in a bank. The public and private keys of Alice and Bob are:

- Alice's public key = (7, 187)
- Alice's private key = (23, 187)
- Bob's public key = (7, 143)
- Bob's private key = (103, 143)

- a) What is the highest possible value for the key combination if Alice wants to use a single encrypted message to send the key combination to Bob? (1 pt)

The key combination should not be higher than 142 in any case.

- b) If the key combination is 101, calculate the encrypted number Alice should send to Bob (2 pt)

$$C = 101^7 \bmod 143 = 62$$

Once the security deposit box is opened, Bob wants to send Alice, using RSA, the amount of money he found inside.

- c) If Bob found 8 million, calculate the message he should be sending to Alicia (1 pt)

$$C = 8^7 \bmod 187 = 134$$

Charles, a bank thief, was able to intercept the message Bob sent to Alicia. Carlos knows both their public keys but doesn't have access to their private ones.

- d) Explain how Carlos would be able to obtain Alicia's private key and make the necessary calculations to obtain it. (2 pt)

Carlos should factor 187 which is 11×17 . Then he would calculate $\phi(187) = 10 \cdot 16 = 160$. From there he would use AEE to calculate $\text{inv}(7, 160)$

T	Q			
0	-	160	1	0
1	-	7	0	1
2	22	6	1	-22
3	1	1	-1	23
4	6	0		